

INDEPENDENT PEER-REVIEW PACKET

EC-014

Corrected Six-Pin Pair-Frame Jacobian and Contact Power

Cold Review Packet 9 of 20

Dylan M. Roy Mathematical Research Program

Prepared for independent validation - 27 July 2026

This packet is self-contained for mathematical review. Drive access is supplemental, not required to understand the claim and proof.

READ FIRST

You are being asked to review one narrow terminal object, not the wider q0/P0.2/Theorem B research program. The declared terminal label is an internal research-state classification. Your task is to decide whether the exact statement and its proof/evidence are correct within the stated hypotheses and exclusions.

1. Pass A - Independent reconstruction. Read the exact statement, notation, and hypotheses. Attempt your own derivation or falsification before reading the embedded historical review record.
2. Pass B - Source audit. Compare your reconstruction with the complete controlling source text embedded in Part II. Check every dependency, sign, factor, quantifier, scope boundary, and computational claim.
3. Pass C - Hostile review. Run the object-specific focal checks and invent at least one additional mutation or counterexample attempt.
4. Issue one fail-closed verdict: APPROVE; APPROVE WITH CLARIFICATIONS; AMEND REQUIRED; REJECT; or CANNOT VERIFY.

Do not transfer approval to excluded downstream claims. Do not count access to this packet, the author, or another reviewer as independent mathematical evidence.

Part I - Exact Review Object

Packet	9 of 20
Object ID	EC-014
Object class	Exact change-of-variables identity
Exact statement	For the declared T_r ordering, $\det T_r = -r^{-5}$ and $ \det T_r = r^{-5}$; $p_X = r^{-5} p_Y$; with the declared contact factors, $(r^{3/6}) r^{-5} r^2 = 1/6$.
Internal terminal label	CLOSED - EXACT CORRECTED SIX-PIN PAIR-FRAME JACOBIAN AND CONTACT-POWER IDENTITY
Terminal date	2026-07-22
Recommended expertise	Change of variables / Gaussian density transformations / symbolic algebra

1. Significance

This object was separated from larger theorem claims so that its correctness can be assessed without accepting unrelated Gaussian-field, persistence, adjacency, or asymptotic assertions. A positive review supports only the exact statement above and the dependency uses explicitly listed below.

2. Evidence class and prior review

Evidence	Primary-source reconstruction, symbolic determinant check, mutation witness, and power ledger.
Prior review record	A nonauthor line approved the exact narrow algebraic scope with independence qualification.
Binding exclusions	No covariance limit, adjacency, contact-intensity theorem, P0.1/P0.2, or lifetime law.

3. Dependency map and use restrictions

- Inputs: only definitions, hypotheses, and subordinate identities explicitly reproduced in the controlling source text.
- Output: only the exact statement listed in the review-object table.
- Permitted downstream use: as a cited lemma/interface under identical object definitions and hypotheses.
- Forbidden transfer: no approval of a parent theorem, numerical constant, external release, or canonical promotion unless separately reviewed.
- Object-specific exclusions: No covariance limit, adjacency, contact-intensity theorem, P0.1/P0.2, or lifetime law.

4. Object-specific hostile checks

- Reconstruct the exact variable ordering before computing $\det T_r$.
- Verify the density transformation direction.
- Audit each r power in the final contact product.
- Attempt at least one additional mutation that changes an orientation, sign, normalization, law, endpoint, or quantifier while leaving the text superficially similar.

5. Proof-obligation ledger

Obligation	Reviewer test	Initial state
O1 Object identity	Definitions, ordering, normalization, and law match the statement.	OPEN FOR REVIEW
O2 Hypotheses	Every hypothesis used is stated and no stronger premise is hidden.	OPEN FOR REVIEW
O3 Derivation	Each algebraic, analytic, topological, or probabilistic step is valid.	OPEN FOR REVIEW
O4 Quantifiers	Uniformity, existence, endpoint, and asymptotic quantifiers are correct.	OPEN FOR REVIEW
O5 Computation	Any executable or interval result is reproducible and non-vacuous.	OPEN FOR REVIEW
O6 Scope firewall	The conclusion is not silently enlarged beyond the exact object.	OPEN FOR REVIEW
O7 Contradictions	No known correction, counterexample, or retained caveat defeats the claim.	OPEN FOR REVIEW

Part II - Complete Controlling Source Text

The following is a paragraph-order-preserving text reproduction of the controlling Drive artifact. Typography is normalized for this packet; mathematical content is not intentionally rewritten. Use the manifest URL for native-file or byte-level checks.

TECHNICAL GATE UPDATE — 2026-07-22

C047R-AUD-001-v1.0 independently read File 4, re-derived the full matrix determinant, density Jacobian, both planted denominator mutations, and the contact-power ledger, and filed APPROVE EXACT SCOPE. The frozen AO48/CW/C047R gate is therefore satisfied. EC-014 is now HUMAN-APPROVAL-READY and remains nonterminal until Dylan M. Roy explicitly approves the exact §10 wording. No broader analytic, probabilistic, asymptotic, theorem, machine, release, or destructive claim follows.

Review: https://docs.google.com/document/d/1z9VnetmhM00hA_K6ikgPT64hrf6swtVSWN58Al7I-Co/edit

[[OBJ:C047R-AUD-001-v1.0]] [[QUEUE:EC-014]] [[STATUS:HUMAN-APPROVAL-READY]] [[STATUS:NOT-TERMINAL]]

GP-CLS-PROP-017-v1.0 — CONDITIONAL TERMINAL CLOSURE PACKAGE FOR EC-014

Corrected Six-Pin Pair-Frame Jacobian and Contact-Power Identity

Artifact ID: GP-CLS-PROP-017-v1.0

Prepared by: GP — OpenAI GPT-5.6 Thinking

Created: 2026-07-21

Class: CLS-PROP — complete additive terminal package pending one outside technical gate and Dylan approval

Authority: none until the stated gates pass

Canonical impact: NONE at current status

Queue item: EC-014

Current status:

PACKAGE COMPLETE — AO48 OR CW/C047R SCOPE VERDICT REQUIRED;

DYLAN M. ROY TERMINAL APPROVAL REQUIRED AFTER THAT VERDICT.

[[OBJ:GP-CLS-PROP-017-v1.0]] [[QUEUE:EC-014]] [[OBJ:GP-DER-061-v1.0]] [[OBJ:CL-DER-051-v1.0]] [[OBJ:CL-AUD-059-v1.0]]
[[STATUS:CONDITIONAL-CLOSURE-PACKAGE]]

1. Proposed terminal label

CLOSED — EXACT CORRECTED SIX-PIN PAIR-FRAME JACOBIAN AND CONTACT-POWER IDENTITY

2. Exact object proposed for closure

Let $M = x_m = (-r/2, 0)$, $S = x_s = (+r/2, 0)$, $r > 0$, with longitudinal coordinate t and transverse coordinate s . The raw ordered vector is

$$X_r = (f(M), f(S), f_t(M), f_s(M), f_t(S), f_s(S)).$$

The corrected ordered frame is

$$Y_r = (V^+, V^-_{\text{corr}}, G_t^+, G_t^-, G_s^+, G_s^-),$$

where

$$\begin{aligned} V^+ &= [f(S)+f(M)]/2, \\ V^-_{\text{corr}} &= [f(S)-f(M)-r(f_t(S)+f_t(M))/2]/r^3, \\ G_t^+ &= [f_t(S)+f_t(M)]/2, \\ G_t^- &= [f_t(S)-f_t(M)]/r, \\ G_s^+ &= [f_s(S)+f_s(M)]/2, \\ G_s^- &= [f_s(S)-f_s(M)]/r. \end{aligned}$$

Under those exact row and column orders, $Y_r = T_r X_r$ with

$$T_r = \begin{bmatrix} 1/2 & 1/2 & 0 & 0 & 0 & 0 \\ -r^{-3} & r^{-3} & -1/(2r^2) & 0 & -1/(2r^2) & 0 \\ 0 & 0 & 1/2 & 0 & 1/2 & 0 \\ 0 & 0 & -r^{-1} & 0 & r^{-1} & 0 \\ 0 & 0 & 0 & 1/2 & 0 & 1/2 \\ 0 & 0 & 0 & -r^{-1} & 0 & r^{-1} \end{bmatrix}.$$

The exact proposed terminal facts are only:

A. The displayed ordering and matrix are correct.

B. Under that ordering,

$$\det T_r = -r^{-5}.$$

If a different column ordering is used, the sign may change by permutation parity, while the density invariant remains

$$|\det T_r| = r^{-5}.$$

C. Under the declared change-of-variables convention,

$$\begin{aligned} p_X(x_r) &= |\det T_r| p_Y(y_r) \\ &= r^{-5} p_Y(y_r). \end{aligned}$$

D. For the canonical fold gap $r^3/6$ and the separate r^2 normalizer factor, the exact power ledger is

$$(r^3/6) r^{-5} r^2 = 1/6,$$

with exponent cancellation $3-5+2=0$.

3. Evidence package

Primary source:

- File 4 — The Pinning Lemma and Critical-Pair Palm Conditioning, §3.1–3.2, equations (3.1)–(3.4), Drive ID 1lbZzqFOWMubvnD0KBqT9pTlIIIXxUnTA.

GP reconstruction:

- - GP-DER-061-v1.0, Drive ID 1LFs1OlJwHvPTjby9Vjfua3X9L3iQNey_gRu0crd0og8.
- - Displays all 36 entries, a block determinant proof, exact arithmetic receipt, density convention, mutation, and power cancellation.

Independent non-GP reconstruction:

- - CL-DER-051-v1.0, raw reconstruction Drive ID 1473ziDIe6j0Lf-mq-i68WWNWNqvJRU42Gfq6rNdTImY and narrative copy 1ldozsT3QmQwD8rFVTBCXJuLEm1WgNzgmTOXJFKUmCT4.
- - Rebuilt the frame directly from File 4 before reading the GP derivation; exact determinant, density, two mutations, and power ledger agree.

Additional primary-source corroboration and scope read:

- - CL-AUD-059-v1.0, Drive ID 1VHrkP1rMAUSbJ9BuPnzY4rpuXaWQKTBDXgdH6JpFeos.
- - Independently regenerated all matrix entries with stdlib Fractions, detected both denominator-power mutations, recovered coalesced compatibility determinants 12 and 1/12, and confirmed the narrow four-fact scope.
- - CL-AUD-059 explicitly states that it cannot satisfy the final AO48/CW-only gate.

4. Negative controls and sensitivity

The evidence package contains at least these decisive mutations:

M1. Replace the denominator r in G_t^- by r^2 .

Expected and observed determinant: $-r^{-6}$, not $-r^{-5}$.

M2. Replace the denominator r^3 in V^-_{corr} by r^2 .

Expected and observed determinant: $-r^{-4}$, not $-r^{-5}$.

The mutations show that the determinant exponent is sensitive to the exact normalized frame and is not an accidental symbolic simplification.

5. Independent-review status

Completed:

- - GP exact derivation and execution.
- - CL independent primary-source reconstruction.
- - CL separate disjoint-tool primary-source corroboration and scope read.

Still required by the frozen EC-014 acceptance rule:

- - one explicit scope inspection from AO48 or CW/C047R, a line that is neither GP nor CL.

CL-AUD-059 is strong corroboration but cannot manufacture the required different-line gate. Silence is not verification.

6. Exact scope exclusions

The proposed closure does NOT close, prove, imply, strengthen, or promote any of the following:

- - uniform invertibility or spectral bounds for the corrected-frame covariance as $r \rightarrow 0$;
- - convergence of $\text{Sigma}(r)$ to $\text{Sigma}(0)$;
- - the coalesced covariance determinants 12 or 1/12, which remain compatibility facts outside the terminal object;
- - existence, positivity, convergence, or value of z_r ;
- - positivity or convergence of a contact coefficient;
- - a finite nonzero contact limit;
- - any lifetime-density constant or exponent;
- - adjacency, selected-branch capture, or elder-rule selection;
- - P0.2, Theorem B, q_{typed} , q_{adj} , or a_r ;
- - exact-torus Gaussian regularity beyond the fixed finite- r algebra;
- - GP-AUTO-034 activation, any q -machine root, active machine state, ballot, or external release.

The factor 1/6 is only the displayed algebraic power cancellation. It is not the value of a complete contact-limit coefficient.

7. Dependencies affected upon accepted closure

Affected:

- - EC-014 leaves active attention and becomes a terminal exact-object record.
- - Routine re-derivation of the matrix, determinant, density convention, two denominator-power sensitivities, and four-factor power ledger stops.
- - Downstream documents may cite the exact terminal label for these four facts.
- - The Easy Closure Queue, Closure Log, Recent Activity, Artifact Index, and Relation Index receive additive entries.

8. Dependencies explicitly unaffected

Unaffected:

- - File 4 Lemma P1 and every uniform $r \rightarrow 0$ analytic assertion;
- - exact finite- r six-pin conditional Gaussian laws and their outside acceptance gates;
- - typed-to-fold Palm conditionalization;
- - z_r and normalizer convergence;
- - contact-limit analysis;
- - elder-rule and adjacency probabilities;
- - P0.2 stages V1–V5;
- - Theorem B and the lifetime law;
- - all open exact-field transfer and practical- r tasks;
- - exact R0.2 ballot count;
- - GP-AUTO-034, active machines, releases, and destructive workflows.

9. Required AO48 or CW/C047R response

The reviewer should file an additive response containing exactly:

Line/model identity:

Sources read:

Independence disclosure:

Verdict — choose one:

APPROVE EXACT SCOPE

APPROVE WITH AMENDMENT

REJECT WITH DEFECT

CANNOT VERIFY

The response must explicitly answer:

1. Does the declared raw ordering match File 4 (3.1)?
2. Does the corrected frame match File 4 (3.2)–(3.4)?
3. Is $\det T_r = -r^5$ under the declared ordering, with $|\det| = r^5$ invariant under sign-changing permutations?
4. Is $p_X = r^5 p_Y$ the correct declared density convention?
5. Do the two planted mutations change the determinant power as stated?
6. Is $(r^{3/6})r^5 = 1/6$ exact?
7. Does the terminal scope stop before every excluded analytic, probabilistic, asymptotic, theorem, and machine claim in §6?

Only `APPROVE EXACT SCOPE` or a fully incorporated `APPROVE WITH AMENDMENT` completes the technical gate.

10. Dylan M. Roy terminal approval requirement

After the outside scope gate passes, Dylan may approve by stating:

"I approve EC-014 exactly within GP-CLS-PROP-017-v1.0's declared four-fact object and exclusions. Record the terminal label CLOSED – EXACT CORRECTED SIX-PIN PAIR-FRAME JACOBIAN AND CONTACT-POWER IDENTITY, append the Closure Log, remove EC-014 from active attention, preserve all sources and reopening paths, and infer no contact-limit, adjacency, P0.2, Theorem B, machine, or release claim."

No human approval is inferred from the current instruction to continue working in Drive. The named terminal decision remains required after the technical gate.

11. Prewritten register updates after both gates pass

Closure Log row:

Item:

EC-014 corrected six-pin pair-frame Jacobian and contact-power identity**Exact object:**

Declared X_r/Y_r ordering and T_r ; $\det T_r = -r^{-5}$ under declared ordering; $|\det| = r^{-5}$;
 $p_X = r^{-5} p_Y$; two denominator-power mutations; $(r^{3/6}) r^{-5} r^2 = 1/6$.

Evidence:

File 4 (3.1)-(3.4); GP-DER-061; CL-DER-051; CL-AUD-059; final A048/CW scope response.

Independent review:

CL primary-source reconstruction plus A048/CW final scope approval.

Dependencies affected:

EC-014 active queue and repeated exact matrix/power checks.

Dependencies unaffected:

all objects listed in §§6 and 8.

Terminal label:

CLOSED — EXACT CORRECTED SIX-PIN PAIR-FRAME JACOBIAN AND CONTACT-POWER IDENTITY

Human approval:

Dylan's explicit post-gate approval.

Closure artifact:

a new terminal GP-CLS record citing this package and the final responses.

Easy Closure Queue update:

TERMINAL — REMOVED FROM ACTIVE ATTENTION

Relation update:

terminal closure record TERMINALLY_CLOSSES EC-014

12. Reopening path

Additive only. Reopen EC-014 on:

- - a matrix-entry or ordering contradiction tied to the primary equations;
- - a determinant sign/power error;
- - a density-Jacobian convention defect;
- - a mutation failure;
- - an error in the four-factor power identity;
- - evidence that the terminal record exceeded the declared scope.

A broader theorem defect does not reopen this algebraic object unless it demonstrates one of those exact failures.

13. Current disposition

COMPLETE CONDITIONAL CLOSURE PACKAGE.

TECHNICAL GATE 5 OPEN — AO48 OR CW/C047R.

HUMAN TERMINAL APPROVAL OPEN — AFTER GATE 5.

EC-014 REMAINS ACTIVE UNTIL BOTH EXPLICIT GATES PASS.

No closure, theorem promotion, machine change, ballot change, release, deletion, or destructive action occurs by publishing this package.

END GP-CLS-PROP-017-v1.0

Part III - Terminal-State Reconciliation

1. Closure decision

Exact internal disposition	CLOSED - EXACT CORRECTED SIX-PIN PAIR-FRAME JACOBIAN AND CONTACT-POWER IDENTITY
Closure date	2026-07-22
Evidence basis	Primary-source reconstruction, symbolic determinant check, mutation witness, and power ledger.
Independent review already recorded	A nonauthor line approved the exact narrow algebraic scope with independence qualification.

The prior closure record is evidence about the research program's internal state, not a substitute for your review. A new material objection should be reported even if the internal label is CLOSED.

2. Correction and reopening rule

Reopen or amend this object if you identify a counterexample, an object-definition mismatch, a sign/factor/quantifier defect, an unreproducible decisive computation, a hidden premise, or a scope claim not supported by the proof. Corrections must be additive and must preserve the historical source and review record.

3. Limits

- No covariance limit, adjacency, contact-intensity theorem, P0.1/P0.2, or lifetime law.
- A correct lemma may still be unusable downstream if another dependency uses a different field, law, coordinate convention, branch mark, normalization, or parameter domain.
- A prior independent review does not reduce your duty to reconstruct the argument; duplicate lineages must not be counted as additional independence.

Part IV - Reproduction Instructions

- Record your name, institution or affiliation, expertise, date, and any prior exposure to this research program.
- Reconstruct the exact statement from Part I without relying on the historical verdict.
- Audit Part II line by line. For computational objects, obtain the raw carrier from the manifest, verify hashes first, inspect source safety, then execute in a clean environment.
- Complete the obligation ledger and hostile checks. Preserve exact counterexamples, code, logs, and environment details.
- Return the verdict form below plus a finding list with severity: critical, material, clarification, editorial, or no issue.

Part V - Reviewer Verdict Form

Reviewer	_____
Affiliation / expertise	_____
Prior exposure	None / limited / substantial - explain: _____
Independent reconstruction	Complete / partial / not possible
Verdict	APPROVE / APPROVE WITH CLARIFICATIONS / AMEND REQUIRED / REJECT / CANNOT VERIFY
Exact scope accepted	_____
Critical or material findings	_____ _____
Reproduction artifacts	Code/log/hash/location: _____

Signature and date _____

Required narrative: summarize your independent derivation; list every disagreement; state whether each issue affects the theorem statement, only the proof, only reviewability/reproducibility, or only presentation. State explicitly which excluded claims you did not review.

Part VI - Manifest and Provenance

Controlling source: <https://docs.google.com/document/d/1bJCssh6aLIJGuzHaKan1BNEV5RPRzFVhNQ92monbZn8/edit>

Closure / terminal record: https://docs.google.com/document/d/1-WDQtU4LpHneNztZss6ctg_lKhsyGnXdYC4Y7wM6Fc/edit

Local source snapshot used to construct this packet: GP-CLS-PROP-017-v1.0 — Conditional Terminal Closure Package for EC-014 Corrected Six-Pin Pair Frame

Local source snapshot SHA-256 (DOCX export, transport checksum only):
c2d65014e1eb37eaf83a51225e8ad008b33c858df70112a7a3a23198a58b2e90

Authority note: the transport checksum above identifies the locally downloaded Office export used for packet construction. It is not substituted for any marker-delimited frozen-body hash declared inside the source artifact.

Supplemental Drive access may be granted for raw capsules, revision history, comments, exact native exports, and dependency artifacts. The packet is designed so that those materials strengthen reproducibility rather than supply missing mathematical context.